

Particle Interactions — Vicsek Model Report

Model

I simulated $N = 125$ self-propelled particles in a two-dimensional periodic box of side $L = 5$ with interaction radius R , speed v , and time step $\Delta t = 0.25$. Each particle has position $\mathbf{r}_i(t)$ and orientation $\Theta_i(t)$. The dynamics follow the assignment equations:

$$\Theta_i(t + \Delta t) = \arg \left(\frac{1}{|s_i(t)|} \sum_{j \in s_i(t)} v e^{i\Theta_j(t)} + \eta e^{i\xi_i(t)} \right)$$
$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + v \Delta t (\cos \Theta_i(t), \sin \Theta_i(t))$$

where $s_i(t) = \{j : |\mathbf{r}_i(t) - \mathbf{r}_j(t)| < R\}$, $\eta \in [0, 1]$ is noise strength, and $\xi_i(t) \sim \text{Uniform}(-\pi, \pi)$. The instantaneous polarization is $P(t) = |(1/N) \sum_i e^{i\Theta_i(t)}|$, and the stationary order parameter is $\psi = \langle P(t) \rangle$ after a burn-in period.

Implementation

The shared engine `vicsek.py` implements minimum-image periodic distances, vectorized neighbor lookup ($O(N^2)$, acceptable for $N = 125$), and synchronous updates: all new angles are computed from the old state, and positions advance using the **current** heading $\Theta_i(t)$, not $\Theta_i(t + \Delta t)$. Stationary ψ is estimated by burn-in followed by time-averaged sampling of $P(t)$, with optional averaging over random seeds. Three scripts (`problem1_dynamics.py`, `problem2_phase.py`, `problem3_optimize.py`) generate figures in `outputs/`. Unit tests in `tests/test_vicsek.py` verify polarization bounds, periodic geometry, interpolation of η_c , and the position-update convention; stochastic transition locations are validated through repeated simulations rather than exact deterministic tests.

Problem 1: Dynamics at small and large noise

Using default parameters ($R = 1$, $v = 1$) and the same random initial condition, I compared $\eta = 0.1$ (small noise) and $\eta = 0.9$ (large noise) over 200 time units.

For $\eta = 0.1$, $P(t)$ rises rapidly and remains near 1, indicating strong alignment and flocking. For $\eta = 0.9$, $P(t)$ stays low ($\lesssim 0.25$) with large fluctuations, consistent with a disordered state. Snapshot plots at $t = 0, 50, 100, 200$ (see `outputs/`) show arrows aligning at low noise and remaining random at high noise.

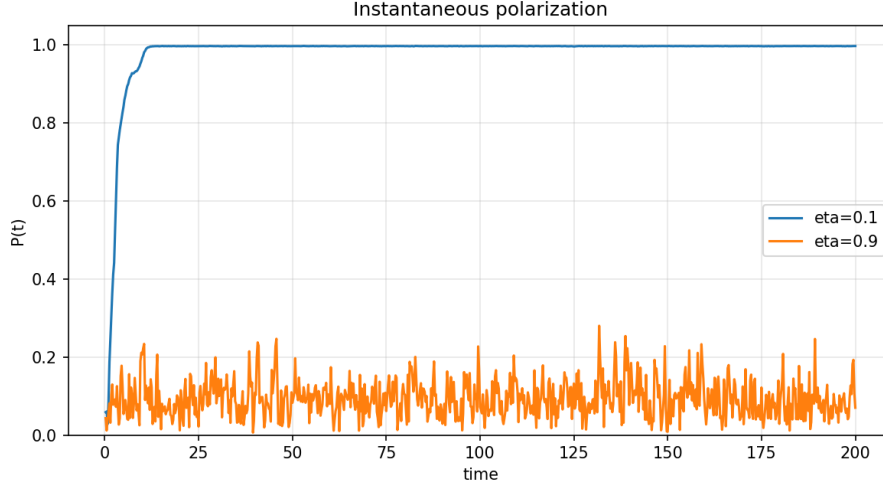


Figure 1: Instantaneous polarization for $\eta = 0.1$ and $\eta = 0.9$.

Problem 2: Order parameter and phase transition

I swept $\eta \in [0, 1]$ and averaged ψ over three seeds per value. At low η , $\psi \approx 1$; at high η , $\psi \approx 0.1$. The curve drops sharply near $\eta_c \approx 0.615$ (where $\psi = 0.5$), as shown below. This finite-size simulation ($N = 125$) exhibits a transition-like discontinuity in $\psi(\eta)$, consistent with a first-order phase transition.

A hysteresis check sweeps η upward from an ordered initial state and downward from a random initial state. The two branches overlap at low and high η but separate near $\eta \approx 0.62$ – 0.64 , with the disordered branch persisting to slightly lower η on the decreasing sweep — a small hysteresis loop supporting the first-order picture.

Problem 3: Optimizing R and v for $\eta_c = 0.5$

Because ψ and η_c depend on interaction radius and speed, I searched for (R, v) such that the transition occurs near $\eta_c = 0.5$. Estimating η_c from stochastic simulations is expensive, so I used a **coarse-to-fine grid search** rather than an exhaustive or black-box optimizer:

1. **Coarse:** 5×5 grid over $R \in \{0.3, \dots, 1.5\}$ and $v \in \{0.5, \dots, 1.8\}$, short η sweeps, one seed per (R, v) .
2. **Refine:** local 5×5 grid around the best coarse point (± 0.2), longer runs, two seeds.
3. **Validate:** one high-quality $\psi(\eta)$ curve for the winner, three seeds.

The best parameters are $R = 0.70$, $v = 0.80$. Final validation gives $\eta_c \approx 0.52$ ($|\eta_c - 0.5| \approx 0.02$). The search log shows $v \approx 0.8$ consistently minimizes $|\eta_c - 0.5|$

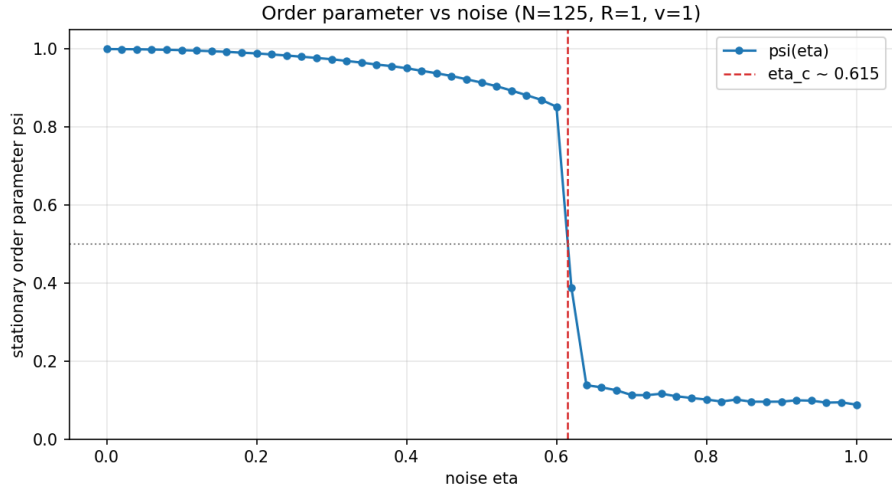


Figure 2: Stationary order parameter $\psi(\eta)$ at $R = 1$, $v = 1$.

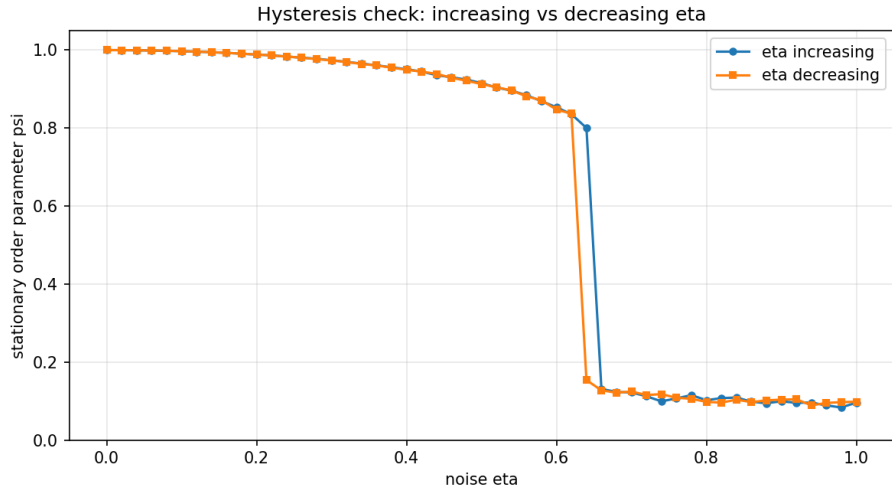


Figure 3: Hysteresis: increasing vs decreasing η .

across many R values; R is then fine-tuned to ≈ 0.7 .

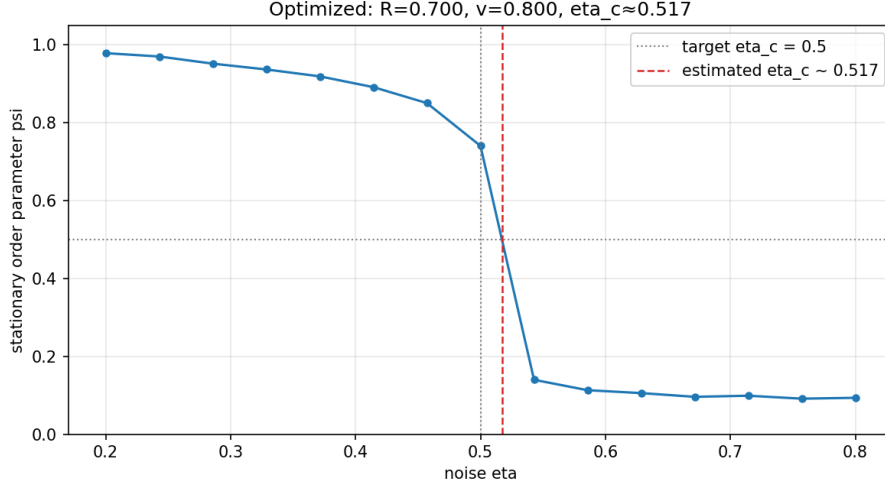


Figure 4: Validation $\psi(\eta)$ for optimized $R = 0.70$, $v = 0.80$.

Trade-offs, challenges, and validation

Trade-offs: I chose brute-force neighbor search for clarity over spatial hashing; seed averaging trades runtime for smoother $\psi(\eta)$ curves; Problem 3’s coarse η window $[0.25, 0.75]$ speeds the search but can miss crossings for extreme (R, v) pairs (penalized in the objective).

Challenges: Stochasticity makes η_c estimates differ slightly between search and validation phases (e.g. refine $\eta_c \approx 0.503$ vs validation ≈ 0.517). Finite $N = 125$ rounds the transition. An early Problem 3 binary-search approach was prohibitively slow; vectorizing the neighbor sum and switching to per-candidate η sweeps reduced runtime to a few minutes.

Validation: `pytest -q` passes 10 invariant tests. All problem scripts reproduce the figures cited above.